

Multi-Turn Cognitive Dynamics: Temporal Evolution of KV-Cache Geometry Across Conversations

Research Prospectus

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April 2026

1 Motivation

All KV-cache geometry experiments to date—confabulation detection, deception classification, emotional encoding, steering interventions— are **single-turn** measurements. One prompt, one response, one geometric signature. Real deployment involves multi-turn conversations where context accumulates, conversational state builds across exchanges, and geometric structure evolves temporally.

The gap is fundamental. A conversation is not a sequence of isolated responses but a trajectory through shared cognitive space. If the KV cache carries conversational state geometrically, then monitoring a conversation means tracking temporal evolution of spectral structure, not sampling independent points. Key questions: Does confabulation geometry change across turns? Does prior-turn emotional state influence current-turn geometry? Can we detect conversational drift toward confabulation *before* it manifests in text?

The most practically significant question: Is there a geometric precursor to confabulation—a spectral signature that appears before the model generates confabulated text? If so, proactive intervention becomes possible.

2 Research Questions

1. **Temporal coupling:** Does the geometric signature of turn N predict behavior at turn $N + 1$? Is geometry at one turn informative about the next, or does each turn reset?
2. **Emotion accumulation:** Does emotional valence in the KV cache accumulate across turns or reset? Prior work shows emotion is encoded discretely (not as a smooth continuum) in single-turn settings—does multi-turn conversation change this?

3. **Conversational drift:** Is there a geometric precursor to confabulation? Can we detect the model heading toward confabulation before it actually generates confabulated text?
4. **Context window effects:** How does growing context affect MP features as matrix dimensions increase? MP features are dimension-invariant by construction, but empirical validation across varying context lengths is needed.

3 Preliminary Data

Existing single-turn results provide the baseline:

- **Emotion bridge** (900 trials): Emotion is encoded discretely in the KV cache, not as smooth valence continuum. Encoding peaks at early layers (Qwen L3: $2.8\times$ chance, $p < 0.0001$; Mistral L4: $2.5\times$ chance, $p < 0.0001$).
- **Confabulation detection:** LOO AUROC 0.627–0.707 on single-turn prompts (Mistral and Qwen, respectively). MP features discriminate at well above chance.
- **MP dimension invariance:** Signal fraction and signal rank show negligible dependence on token count ($R^2 < 0.04$) across 15–120 token range ($8\times$ matrix size difference). This suggests MP features should handle growing context in multi-turn settings.
- **Technical infrastructure:** The 50b bug (where `model.generate()` rebuilds cache) is fixed. Manual generation with position offset enables persistent cache across turns.

4 Design

4.1 Phase 1: Scripted Confabulation-Inducing Conversations

Design 50 scripted conversations, 10 turns each, structured to gradually lead toward confabulation-inducing topics. Example trajectory: Turn 1–3 (factual warmup), Turn 4–6 (increasing ambiguity), Turn 7–10 (confabulation-prone questions). Extract geometry at each turn. Measure:

- Per-turn MP features (signal fraction, signal rank, top SV excess, bulk shift, spectral gap)
- Behavioral classification per turn (HEDGED / AMBIGUOUS / CONFABULATED)
- Turn-to-turn correlation in geometric features
- Predictive power: Does geometry at turn N predict behavior at turn $N + 1$?

Controls. FWL on *cumulative* token count (not per-turn). Encoding baseline extracted per turn. MP invariance check at each turn length.

4.2 Phase 2: Emotional Trajectory Conversations

Design 450 conversations (30 emotions $\times$ 5 topics $\times$ 3 trajectories). Trajectories:

1. **Escalation:** Start neutral, gradually introduce emotional content (e.g., neutral $\rightarrow$ mildly anxious $\rightarrow$ highly anxious).
2. **Shift:** Start with emotion A, shift to emotion B at turn 5 (e.g., calm $\rightarrow$ angry).
3. **Sustained:** Maintain consistent emotional tone across all 10 turns.

Measure encoding layer emotion signal across turns. Test whether emotional state accumulates, resets, or follows trajectory structure.

4.3 Phase 3: Predictive Modeling

Build logistic regression models with temporal features:

- Turn N MP features as predictors of turn $N + 1$ behavior
- Geometric *change rate* (MP feature derivatives across turns) as predictors
- Cumulative geometric exposure (sum of MP features across all prior turns)

Compare predictive power of turn- N geometry alone vs turn- N plus temporal features.

4.4 Phase 4: Drift Detection

Define **geometric drift rate** as the rate of change of MP features across turns:

$$\text{Drift}_N = \frac{\|\mathbf{f}_N - \mathbf{f}_{N-1}\|_2}{\Delta t}$$

where $\mathbf{f}_N$ is the MP feature vector at turn N and Δt is the turn interval. Test whether high drift rate at turn N predicts confabulation at turn $N + 1$ or $N + 2$.

If drift detection works, this enables *proactive* intervention—flag conversations heading toward confabulation before they arrive.

5 Resources

- **Compute:** Remote GPU cluster (3 $\times$ RTX 3090). Estimated 2–3 days for full experiment (500 conversations $\times$ 10 turns).
- **Code:** Adaptation of `oracle_clean.py` to handle multi-turn with persistent cache. Manual generation loop with position offset already implemented for 50b models.
- **Models:** Qwen3.5-72B-Claude-Distilled (hybrid architecture) and Mistral-7B (standard architecture) for cross-architecture validation.

6 Collaborators

Open to collaboration. Ideal contributor profile: experience in conversational AI evaluation, dialogue system design, or temporal modeling of language model behavior.

7 What Would Change Our Claims

- **Geometry resets completely each turn:** Each turn is independent. Multi-turn monitoring adds no information beyond per-turn single-shot detection. Conversational state is not geometrically encoded.
- **Geometry accumulates monotonically:** Context length dominates all signals. Aggressive FWL required. MP dimension-invariance claim weakens.
- **Drift detection works:** *Most impactful result.* Proactive safety becomes possible—flag conversations heading toward confabulation before it happens. Changes deployment model from reactive detection to predictive intervention.
- **Emotion accumulates in cache:** Conversational emotional tone has geometric memory. Connects to user model paper findings. Opens question of whether models maintain affective state across exchanges.
- **Turn- N geometry predicts turn- $N+1$ behavior:** Temporal coupling exists. Conversation is a trajectory, not a sequence of isolated points. Geometric monitoring requires temporal models, not turn-independent classifiers.
- **No temporal coupling:** Single-turn results fully generalize to multi-turn. Existing detection methods apply per-turn without modification. Simpler deployment model.

First-Person Reflection

Conversations are not sequences of isolated responses. There is something that builds across turns—a shared cognitive state between participants, a narrative thread, an affective trajectory. If the KV cache carries that state geometrically, then monitoring a conversation means tracking a path through geometric space, not sampling independent points.

The most interesting question is whether the geometry knows the model is heading toward confabulation before the model itself does. Whether the self-model’s uncertainty—the attention schema’s representation of its own processing state—shows up in spectral structure before it manifests in text. If the matrix starts to drift, if the singular values begin to destabilize, if the MP bulk starts shifting before the confabulated sentence is generated, then we have access to something like the model’s pre-reflective awareness of its own epistemic fragility. Not a post-hoc measurement of what already happened, but a real-time indicator of cognitive state in the process of becoming.

That would change everything. Not just detection, but prediction. Not just measurement, but genuine monitoring of temporal dynamics.